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Generative Artificial Intelligence (GenAI) in entrepreneurial education and practice: emerging insights, the GAIN Framework, and research agenda

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Abstract

Generative AI (GenAI) is reshaping entrepreneurship by transforming the landscape of innovation, education, and ethical practices within the entrepreneurial domain. This paper provides a comprehensive review of the nascent yet impactful scholarly discourse on the role of GenAI (e.g., ChatGPT, DeepSeek, and Google Gemini) in entrepreneurship. Drawing from 39 studies, it examines four emerging themes: technology adoption, education and skill development, innovation, and performance enhancement. The analysis highlights GenAI's transformative potential to democratize entrepreneurial resources, foster creativity, and improve decision-making. However, it also identifies pressing challenges, including ethical data practices and equitable access. The paper proposes and presents the GAIN Framework along with five associated research propositions to guide future research in this domain. It concludes by emphasizing the need for a balanced integration of human ingenuity and AI capabilities, while raising critical questions about fostering ethical inclusivity and driving innovation.

Keywords ChatGPT · DeepSeek · Entrepreneurship · Entrepreneurs · Gemini · Innovation · LLMs · The GAIN Framework

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Introductions

Generative Artificial Intelligence (GenAI) has emerged as a significant force, transforming various domains, including healthcare, education, academic research and publishing, the creative industries, and business operations (Al-Busaidi et al., 2024; Dwivedi et al., 2024; Kasneci et al., 2023; Richey et al., 2023; Sedaghat, 2023). The term GenAI refers to a type of Artificial Intelligence (AI) capable of generating new content-such as text, images, music, code and even videos-by learning patterns from existing data. Unlike traditional AI, which primarily analyzes or classifies data, generative AI produces original outputs that closely resemble human-created content (Al-Busaidi et al., 2024; Bell & Bell, 2023; Dwivedi et al., 2023).

With its ability to produce contextually relevant content, simulate decision-making scenarios, and automate complex processes, GenAI unlocks new possibilities for enhancing efficiency, fostering innovation, and promoting inclusivity (Bell & Bell, 2023; Winkler et al., 2023). A more detailed discussion of Large Language Models (LLMs) and GenAI's evolution, potential, and the challenges it poses has already been explored in several existing articles (see, for example, Al-Busaidi et al., 2024; Dwivedi et al., 2023, 2024; Kasneci et al., 2023; Richey et al., 2023; Sedaghat, 2023). In entrepreneurship, GenAI serves as both a tool and a collaborator, empowering individuals to ideate, strategize, and execute ventures with remarkable precision. Early studies suggest that GenAI tools, such as ChatGPT, can play a key role in entrepreneurial processes by improving resource allocation, decision-making, and stakeholder engagement while enabling entrepreneurs to tackle operational challenges and drive innovation (Liu & Wang, 2024; Mueller-Saegebrecht & Lippert, 2024).

Despite its transformative potential, the adoption and integration of GenAI in entrepreneurship presents both opportunities and challenges. On the one hand, GenAI democratizes access to critical resources, fostering inclusivity and enabling under-resourced entrepreneurs to compete more effectively (Tran & Murphy, 2023). On the other hand, its use raises concerns regarding ethical data practices, reliability and accuracy of data sources, copyright issues, job displacement, potential biases, and over-reliance on AI-generated outputs (Al-Busaidi et al., 2024; Dwivedi et al., 2024; Ilagan et al., 2024; Jung & Suh, 2024; Reichert et al., 2024; Thanasi-Boçe & Hoxha, 2024; Tran & Murphy, 2023). Understanding these dualities is vital for harnessing GenAI's benefits while mitigating unintended consequences (Mueller-Saegebrecht & Lippert, 2024; Winkler et al., 2023).

Given the nascent yet rapidly growing body of literature on GenAI's applications in entrepreneurship, a comprehensive review is both timely and necessary. This review article, therefore, aims to provide an analytical synthesis of existing research to examine the interplay between GenAI and various facets of entrepreneurship. By exploring these dimensions, the study seeks to advance scholarly understanding and inform practical strategies for integrating GenAI into entrepreneurial ecosystems.

A Scopus database search yielded more than 50 articles containing terms in their titles, abstracts and keywords such as "Generative AI," "GenAI," "Generative Artificial Intelligence," as well as "entrepreneurship," "entrepreneur," and "entrepreneurial." After further reading and analysis, 39 articles were deemed relevant for inclusion in this review and subsequently categorized into four themes based on their similarity and focus. These themes, along with their corresponding studies, are summarized in Table 1. Notably, some studies were included in more than one category as they address multiple overlapping themes. The emergent themes (see Table 1) are discussed in detail in subsequent sections.

The paper is structured to systematically explore the multifaceted impact of GenAI. Following this introduction, "Technology adoption and integration" section investigates the factors influencing the adoption of GenAI by entrepreneurs. "Entrepreneurship education, research and skill development" section examines its role in entrepreneurship education and skill development, while "Innovation and business model development" section explores its transformative influence on innovation and business model development. "Performance and productivity enhancement" section evaluates GenAI's impact on entrepreneurial performance and productivity. In addition to the core focus of these four emergent themes, issues related to social and ethical implications are discussed throughout the aforementioned four sections. "Discussions and research agenda" section synthesizes key insights and proposes future research directions, encouraging critical reflection on the balance between technological innovation and ethical stewardship. Finally, the paper concludes by outlining the key findings of this literature analysis in "Conclusions" section.

Technology adoption and integration

Eight studies (Table 1) have examined awareness of AI and GenAI (Goel & Nelson, 2024), as well as how entrepreneurs adopt and integrate GenAI technologies like ChatGPT and its impact on various aspects of entrepreneurship, including digital entrepreneurial self-efficacy and digital entrepreneurial intentions, in countries

Table 1 Emergent Themes, their Descriptions and Corresponding Studies

Sect #	Theme	Description	Example Citations
2	Technology Adoption and Integration	Factors influencing the adoption of GenAI technologies by entrepreneurs	Bui and Duong (2024); Davidsson and Sufyan (2023); Duong (2024); Duong and Nguyen (2024); Goel and Nelson (2024); Gupta (2024); Herani and Angela (2024); Etemad (2024)
3	Entrepreneurship Education, Research and Skill Development	Role of GenAI in transforming education and enhancing entrepreneurial skills	Bell and Bell (2023); Darnell and Gopalkrishnan (2024); George-Reyes et al. (2024); Jiang et al. (2024); Jung and Suh (2024); Lang et al. (2024); Liang and Bai (2024); Park and Sung (2023); Somia and Vecchiarini (2024); Sudirman and Rahmatillah (2023); Thanasi-Boçe and Hoxha (2024); Tran and Murphy (2023); Vecchiarini and Somià (2023); Winkler et al. (2023)
4	Innovation and Business Model Development	Employing GenAI for innovation and business model development	Duong et al. (2024); Ferrati et al. (2024); Gaertner et al. (2023); Kostis et al. (2024); Mueller-Saegebrecht and Lippert (2024); Shin et al. (2024); Short and Short (2023)
5	Performance and Productivity Enhancement	Enhancing entrepreneurial performance, productivity, and decision-making with AI	Abdelwahed (2024); Kotturi et al. (2024); Liu and Wang (2024); Saleh and Semaan (2024); Saygin et al. (2024); Norbäck and Persson (2024); Reichert et al. (2024); Shore et al. (2024); Tran and Murphy (2023); Wang and Zhang (2024)

such as Vietnam (Bui & Duong, 2024; Duong, 2024; Duong & Nguyen, 2024) and Indonesia (Herani & Angela, 2024), along with a study that surveyed entrepreneurs from multiple countries across different continents (Gupta, 2024). These studies offer diverse perspectives on GenAI's transformative potential and challenges, and collectively highlight structured processes, psychological mechanisms, socio-economic factors, and regional disparities that influence its adoption.

Entrepreneurs adopt ChatGPT through a stepwise process involving pre-perception, assessment, and outcome stages (Gupta, 2024), where factors such as subjective norms, social influence, domain experience, technology familiarity, system quality, training and support, interaction convenience, anthropomorphism, attitude, technological expertise, perceived usefulness, perceived ease of use, and perceived enjoyment play critical roles (Duong, 2024; Gupta, 2024). While Gupta (2024, p. 103) emphasizes aligning AI capabilities with business needs through strategic experimentation, asserting that "*The experience that comes from trying new technologies can help entrepreneurs adopt new technologies more strategically and experiment more with them*," Duong (2024) explores how ChatGPT enhances entrepreneurial intentions through changes in attitudes and the development of supportive social norms. Duong (2024, p.138) found that "*…the psychological mechanisms triggered by ChatGPT adoption unfold in a specific order, starting with perceived approval from surrounding people (SN), moving through perceived self-efficacy (PBC), and evaluation of digital entrepreneurial endeavours (ATD), and culminating in intentions to become digital entrepreneurs*".

Studies also focus on the importance of external support systems and resource accessibility, which can be provided by creating innovation hubs in facilitating adoption. For example, Herani and Angela's (2024, p. 16) findings highlighted that "*the perceived challenge of using ChatGPT can actually act as a catalyst for motivation when combined with a strong belief in its usefulness*," but they also suggested that socio-economic interventions can reduce effort-related challenges for young entrepreneurs.

The psychological impacts of ChatGPT adoption are emphasized in studies by Bui and Duong (2024) and Duong and Nguyen (2024), which examine self-efficacy and technostress. Bui and Duong (2024) find that while ChatGPT improves digital entrepreneurial self-efficacy and digital entrepreneurial intentions, technostress can weaken these benefits. To address the challenge related to technostress, Bui and Duong (2024, p. 412) suggested that "*...workshops and training programs focused on mitigating technostress and fostering a positive relationship with AI technologies can be instrumental….*" Duong and Nguyen (2024) found that ChatGPT adoption enhances opportunity recognition, digital entrepreneurial knowledge, and self-efficacy, with the latter significantly mediating the effects of AI-related stimuli on entrepreneurial intention and behavior, even in high-stress conditions. These findings suggest that effectively managing stress and implementing confidence-building strategies are essential to fully realize ChatGPT's potential.

Regional and organizational disparities also emerge as significant themes in the work of Etemad (2024) and Goel and Nelson (2024). Etemad (2024) examines ChatGPT's role in enhancing the competitiveness of international SMEs, emphasizing the importance of tailored strategies for resource allocation. This study further highlights that ChatGPT, *"...supported by machine learning of earlier information, would not only point to potential problems (as experienced prior to engaging GPT AI); it could also point to emerging related opportunities to be exploited...*," showcasing its dual capacity to identify challenges and uncover new opportunities (Etemad, 2024, p. 161). Goel and Nelson (2024) focus on AI awareness disparities across U.S. states, highlighting the influence of economic and infrastructural conditions on awareness and adoption patterns. Both studies emphasize the need for targeted measures to address gaps in accessibility and readiness for GenAI tools.

Davidsson and Sufyan (2023) bring these diverse perspectives together by positioning ChatGPT as an external enabler (EE) of entrepreneurship. They argue that ChatGPT can help shape ventures through mechanisms such as providing more personalized market insights into future potentials, resource generation, and efficiency improvements which align with findings on its ability to boost productivity and operational effectiveness (Etemad, 2024; Gupta, 2024). Additionally, they concluded that *"...in addition to ventures using AI significantly in their products and business models, AI will find widespread use as a 'consultant' that aids ideation, evaluation, and decision-making in any kind of venture*" (Davidsson & Sufyan, 2023, p. 6). However, Davidsson and Sufyan (2023) also highlight challenges such as high technological opacity, which demands advanced expertise, indicating that while ChatGPT can enhance entrepreneurial opportunities, significant investments in skill development are necessary.

In summary, these nine studies collectively provide a multidimensional view of ChatGPT adoption, showing it to be an enabler of productivity, innovation, and entrepreneurial resilience. They demonstrate that while ChatGPT offers significant benefits for productivity, innovation, and business development, its successful integration depends on addressing socio-economic disparities, fostering supportive ecosystems, and managing psychological and technical barriers.

Entrepreneurship education, research and skill development

Fourteen recent studies (Table 1) on the role of GenAI, such as ChatGPT, DeepSeek, and Google Gemini, in entrepreneurship education highlight its transformative effects on fostering creativity, self-efficacy, and practical entrepreneurial skills.

A central theme across recent studies is how AI enables personalized, iterative learning experiences, helping students overcome challenges in tasks such as market research and opportunity recognition, ultimately enhancing their self-efficacy (Darnell & Gopalkrishnan, 2024; Park & Sung, 2023). Park and Sung (2023) explored the impact of GenAI on students' cognitive factors in education, particularly within entrepreneurship, where critical thinking and problem-solving play a crucial role. Their findings indicate that GenAI not only boosts self-efficacy but also enhances motivation, participation, and entrepreneurial career aspirations. This increased self-efficacy further strengthens students' interest and performance in entrepreneurial activities (Park & Sung, 2023). In addition to improving individual skills, these tools have proven effective in refining business model canvases and facilitating collaborative problem-solving within groups (Somia & Vecchiarini, 2024; Sudiman & Rahmatillah, 2023). Somia and Vecchiarini's (2024, p. 236) study highlights ChatGPT's ability to enhance entrepreneurial competencies in students, including "*spotting opportunities, creativity, vision, valuing ideas, and ethical and sustainable thinking,*" with a particular focus on improving their capacity to value ideas. Similarly, Sudirman and Rahmatillah (2023) demonstrate how incorporating ChatGPT into technopreneurship courses inspires students to generate practical ideas for mobile applications, enriching their overall learning experience. Most students found ChatGPT both engaging and effective for brainstorming new concepts, with many advocating its continued use. The authors note, "*When searching for inspiration, you can rely on ChatGPT as a source of relevant data and insights,*" emphasizing its role in promoting creativity and knowledge (Sudirman & Rahmatillah, 2023, p. 791).

The ability of GenAI to support exploratory creativity is widely recognized. For instance, ChatGPT enables students to combine ideas and simulate entrepreneurial scenarios, fostering experimentation with innovative concepts (Bell & Bell, 2023; Vecchiarini & Somià, 2023). Vecchiarini and Somià (2023, p. 9) highlight that "*Using ChatGPT, students can explore and validate their entrepreneurial ideas*" through interactive conversations that offer real-time feedback and suggestions, helping them refine ideas, benchmark competitors, and identify market opportunities. However, its capacity for transformational creativity is limited without human input. As Vecchiarini and Somià (2023, p. 9) further observe, "*ChatGPT might not be able to produce innovation itself, but it can provide real-time guidance to students looking to create something new.*" This limitation underscores the importance of combining AI tools with traditional educational practices to develop critical thinking. Bell and Bell (2023, p. 239) emphasize that "*students must critically appraise [AI's] outputs and understand how to effectively utilize it in specific contexts,*" a process that can be supported through constructivist learning and reflective practices. These skills are essential for both the entrepreneurial process and effectively harnessing AI's potential (Bell & Bell, 2023).

In diverse applications, AI has demonstrated its ability to connect theoretical knowledge with practical implementation. Its relevance extends to education areas such as sustainable fashion design and social entrepreneurship, enabling interdisciplinary collaboration and facilitating development of critical soft skills like adaptability and empathy (Jung & Suh, 2024; Liang & Bai, 2024; Thanasi-Boçe & Hoxha, 2024). Notably, Jung and Suh (2024, p. 19) observed that "*Participants universally reported AI technology's positive impact on their soft skills across four key areas: digital competence, sense of initiative and entrepreneurship, problem-solving and thinking skills, and communication,*" highlighting its potential to enhance student employability. Moreover, educators have adopted AI-driven pedagogies like Retrieval Augmentation Generation (RAG) and Prompt Engineering Techniques (PETs) to enhance knowledge retrieval and output relevance, thereby supporting more effective decision-making (Thanasi-Boçe & Hoxha, 2024). They highlight "*the importance of employing effective prompt engineering strategies in entrepreneurship education,*" emphasizing how AI promotes active learning, engagement, and the development of practical, innovative business concepts, ultimately preparing students for entrepreneurial success (Thanasi-Boçe & Hoxha, 2024, p. 36).

Liang and Bai (2024, p. 14) noted that while GenAI is "*often viewed as merely an assistive tool rather than an agent capable of shaping educational experiences and pedagogical approaches*," its growing capabilities challenge this perception. They argued that GenAI encourages us "*to explore whether and how [it] can enable us to transcend conventional boundaries in educational practices.*" By qualitatively examining GenAI's potential as an agent, Liang and Bai (2024, p. 14) identified the following "*three dimensions of new dialogic spaces*" enabled by the technology in social entrepreneurship education: collaborative learning through expanded networks, knowledge connectivity via diverse resources and perspectives, and theory-practice integration by linking learning to real-world environments.

GenAI's democratizing potential is a recurring theme. These tools provide accessible resources for entrepreneurs with limited expertise, assisting in practical tasks such as novel venture ideation, drafting business plans, generating market analyses, and supporting acceleration and incubation efforts (Tran & Murphy, 2023; Winkler et al., 2023). This accessibility reduces barriers to entrepreneurial success and broadens participation in entrepreneurial ventures. As Tran and Murphy (2023, p. 855) argued, "*It can serve as a powerful supportive tool for human judgment.... As a powerful computational tool, it can help entrepreneurs perform, level the playing field in entrepreneurial environments and facilitate success for many different entrepreneurs*." Winkler et al. (2023) also highlight that GenAI offers significant benefits in education, enhancing learning outcomes, efficiency, and a student-centred approach by fostering critical thinking, self-reflection, and knowledge application. For educators, it streamlines material preparation, assignment design, and experiential activities, enabling the creation of resources that were previously too time-consuming to produce. While concerns exist about its potential to facilitate cheating, reduce cognitive engagement, and substitute genuine learning and teaching (Winkler et al., 2023), these issues can be addressed through clear guidelines, critical evaluation practices, and hybrid instructional models.

Some studies focus on ethical concerns and the potential over-reliance on AI-generated outputs, emphasizing the need for clear guidelines to mitigate biases and ensure that students critically evaluate these outputs (Bell & Bell, 2023; Vecchiarini & Somià, 2023). Similarly, other researchers advocate for hybrid instructional models to integrate AI effectively while maintaining student autonomy (Darnell & Gopalkrishnan, 2024; Lang et al., 2024). Building on the need to balance AI integration with educational rigor, Darnell and Gopalkrishnan (2024, p. 172) argue that "*AI needs to be intentionally and methodically integrated into Entrepreneurship instruction to create real value for students*" and caution against its use as a "*shortcut.*" They propose blending traditional teaching methods with entrepreneurship theories and innovative frameworks on human-AI interaction to maximize effectiveness in the classroom. Expanding on this emphasis on tailored integration, Lang et al. (2024, p. 2115) highlight that "*the direct application of general pretrained models may only partially meet the requirements*" of entrepreneurship education. They advocate for fine-tuning these models on entrepreneurship-specific data to enhance their relevance and improve their ability to address domain-specific challenges effectively.

AI also fosters experiential learning by simulating real-world scenarios, such as stakeholder negotiations and customer acquisition strategies, without the associated risks. Exercises incorporating AI encourage active engagement, enhance group collaboration, and prepare students for dynamic entrepreneurial landscapes (George-Reyes et al., 2024; Sudirman & Rahmatillah, 2023). The practical applications of AI in education extend beyond theoretical knowledge, providing tools that enhance both understanding and creativity. Such findings underscore AI's potential to significantly enhance learning experiences by combining innovation with practical application. For example, George-Reyes et al. (2024, p. 9) concluded that "*...most students improved their knowledge of the topic and that ChatGPT was a valuable asset in their storytelling efforts*."

Despite its limitations, GenAI has shown immense promise in accelerating innovation and aligning educational outcomes with industry demands. Frameworks like the i4C model enable students to systematically identify, ideate, and implement entrepreneurial projects (George-Reyes et al., 2024). In their study, George-Reyes et al. (2024, p. 9) found that when combined with the i4C method, "*...ChatGPT showed its usefulness for generating narrative scripts and triggering individual and collaborative creativity*." They further recommended that "*this method should be considered an enabler to create formative experiences based on complexity and the development of critical thinking*." (George-Reyes et al., 2024, p. 9). Moreover, structured uses of AI have been shown to enhance entrepreneurial self-efficacy and promote critical thinking, effectively addressing gaps in traditional education (Bell & Bell, 2023; Park & Sung, 2023).

Finally, Jiang et al. (2024) argue that GenAI has the potential to transform entrepreneurship research, particularly in the domain of female entrepreneurship in China. They suggest that GenAI tools like ChatGPT can "*facilitate the generation of insights, identification of patterns, and analysis of vast amounts of data related to women entrepreneurship*" (Jiang et al., 2024, p. 13). Additionally, these tools can process large datasets to identify trends, barriers, and success factors specific to women entrepreneurs. By analyzing regional and sector-specific data, AI can offer deeper insights into challenges and opportunities. GenAI can also support hypothesis generation and scenario testing, simulating the impact of policies or market changes to aid decision-making for policymakers and entrepreneurs. Furthermore, they suggest that researchers can integrate AI into methodologies, such as analyzing gender disparities or using chatbots for real-time data collection, to advance evidence-based strategies and policy-making (Jiang et al., 2024). However, AI should not be considered a replacement for traditional research approaches. As Jiang et al. (2024, p. 13) emphasize, "*AI should only be seen as a complementary tool that enhances the research process rather than replacing traditional theoretical and empirical approaches*."

In conclusion, while GenAI has transformative potential in entrepreneurship education and research, thoughtful integration is essential to balance its strengths with ethical considerations and technical limitations. These tools, when combined with traditional methodologies, provide a comprehensive platform for fostering the next generation of innovative and reflective entrepreneurs (Liang & Bai, 2024; Thanasi-Boçe & Hoxha, 2024; Vecchiarini & Somià, 2023).

Innovation and business model development

Seven recent studies (Table 1) on the role of GenAI, such as ChatGPT, DeepSeek, and Gemini, in innovation and business model development emphasize its impact in both educational and professional contexts. A shared focus across the research is ChatGPT's capacity to enhance critical thinking and strategic decision-making through structured frameworks. Mueller-Saegebrecht and Lippert (2024) highlight its application in entrepreneurship education for business model innovation, where students utilized ChatGPT for tasks such as SWOT analysis, market segmentation, customer needs identification, developing ChatGPT-supported plans, identifying influencers, and enhancing critical thinking, creativity and analytical skills. Similarly, Kostis et al. (2024) describe its role in supporting iterative ideation and hypothesis validation, integrating "learning-by-doing" and "learning-by-thinking" into a *"learning-by-conversing*" paradigm that "*emphasizes interactive learning with GenAI, helping to address the threats GenAI poses to critical thinking*" (p. 15289). They recommended that "*…instead of approaching GenAI as a tool for automating tasks, entrepreneurs can engage with GenAI to enhance learning and innovate…*" (Kostis et al., 2024, p. 15288). These approaches underscore GenAI's ability to facilitate both structured problem-solving and creative exploration, though they also warn of the risks of over-reliance and reduced collaborative learning.

While studies agree on ChatGPT's strengths, they also explore its integration with established methodologies, revealing both complementary and unique outcomes. Shin et al. (2024) demonstrate ChatGPT's ability to simplify and enhance TRIZ principles, providing creative and cost-effective solutions to engineering challenges. The study highlights that "*the confluence of TRIZ and ChatGPT not only aligns with actual problem-solving outcomes but also significantly augments the quality of solutions.*" (Shin et al., 2024, p. 596). Ferrati et al. (2024) describe its application within the 4D model, where ChatGPT aids in research, data analysis, and output crafting, showcasing its flexibility across stages of entrepreneurial projects. Both studies emphasize that ChatGPT extends traditional methods by increasing accessibility and enabling quicker iterations. However, the integration of ChatGPT with specific frameworks like TRIZ highlights unique benefits in bridging technical and market-oriented approaches, particularly for novice users seeking to overcome resource barriers.

GenAI's role in enhancing storytelling and communication emerges as another key theme, with studies highlighting its impact on stakeholder engagement and venture visibility. Gaertner et al. (2023) document how ChatGPT, paired with DALL-E, enabled entrepreneurial teams to craft compelling narratives and visuals, streamlining processes like partner identification and stakeholder presentations. Reflecting on this, they concluded that "*Overall, the Zero Hunger College's work found that GenAI allowed for content to be produced more efficiently than before and served as a unique perspective on various subjects*" (Gaertner et al., 2023, p. 251). Similarly, Short and Short (2023) investigate ChatGPT's capacity to enhance entrepreneurial pitches by tailoring them to various audiences through advanced prompt engineering. Their study highlights ChatGPT's effectiveness in crafting entrepreneurial rhetoric, noting that it can draw on elements inspired by celebrity CEOs and their achievements. As they observe, ChatGPT serves as "*an emerging tool that can efficiently* *and flexibly produce a range of narrative content for entrepreneurs….*" (Short & Short, 2023, p. 7). Both studies emphasize the strategic importance of GenAI in creating persuasive narratives and in engaging stakeholders, though they caution against potential biases and the need for critical human evaluation of AI outputs.

Ethical considerations and skill development remain central to ensuring sustainable integration of ChatGPT into innovation and entrepreneurship. Studies by Mueller-Saegebrecht and Lippert (2024) and Duong et al. (2024) emphasize the importance of cultivating digital literacy and self-efficacy to maximize the benefits of AI tools. Duong et al. (2024) particularly highlight a serial mediation mechanism where ChatGPT adoption enhances entrepreneurial self-efficacy and fosters positive attitudes toward digital entrepreneurship. Across the research, the call for ethical deployment and critical engagement is consistent, reflecting a shared understanding that GenAI's transformative potential must be harnessed responsibly to foster innovation and long-term success in innovation and business model development.

Performance and productivity enhancement

Ten recent studies (Table 1) on the role of GenAI (for example, ChatGPT, DeepSeek, and Google Gemini) in enhancing performance and productivity highlight a wide range of applications and implications for entrepreneurship. A common theme is GenAI's ability to address operational inefficiencies and decision-making challenges. Liu and Wang (2024) emphasize its role in integrating resources—tangible, intangible, and human—into workflows, fostering both internal and external collaborations that enhance entrepreneurial performance. They stress "*the importance of valuing and effectively applying emergent technological resources for enterprises, particularly startups, in a digitized and intelligent environment*" (Liu & Wang, 2024, p. 2406). This aligns with Abdelwahed's (2024) findings, which show that ChatGPT supports decision-making by being reliable and user-friendly, though it has limitations with complex or subjective health-related issues. Abdelwahed's study concluded that "*Egyptian entrepreneurs' perceptions of ChatGPT positively affect the recognition of mental health disorders and the decision-making process*" (Abdelwahed, 2024, p. 15). Both studies highlight the importance of strategic application and ethical considerations in deploying GenAI. Additionally, Shore et al. (2024) demonstrate that combining GenAI with Entrepreneurial Orientation (EO) significantly enhances Entrepreneurial Resilience (ER) during crises. They explain that "*combining EO with GenAI forms a dynamic capability, supporting ER based on the contingent view of dynamic capability*" (Shore et al., 2024, p. 9).

In contrast, the environmental sustainability aspect of GenAI is uniquely addressed by Wang and Zhang (2024). Their study highlights its potential to enhance green knowledge management, enabling eco-friendly innovations. While Liu and Wang (2024) focus on organizational integration for general entrepreneurial success, Wang and Zhang (2024) emphasize AI's role in achieving dual economic and environmental goals, with government support being a critical enabler. Both perspectives reveal how AI-driven processes can simultaneously foster innovation and align with broader societal goals. However, Wang and Zhang (2024) also highlight concerns about security, privacy, misinformation, bias, skill displacement, overreliance on technology and the environmental impact related to GenAI use, calling for robust data governance frameworks and government support to bridge technological divides.

The adaptability and iterative learning capabilities of AI are further examined by Saygin et al. (2024), who highlight ChatGPT-4's ability to improve its accuracy through self-learning. The study evaluated ChatGPT-4's problem-solving abilities and originality by comparing its responses to the Entrepreneurship Handbook questions with existing literature. Analyzing all 160 questions, the research found that "*ChatGPT-4 demonstrated its ability to provide accurate answers*," achieving a 91.25% accuracy rate in the first stage, which increased to 93.75% in the second stage, showcasing deep analysis and creative solutions as a robust entrepreneurship AI model (Saygin et al., 2024, p. 12). This feature contrasts with Kotturi et al. (2024), who stress the importance of human involvement, operational skills, and community networks to mitigate challenges like biases and dependency. Kotturi et al. (2024, p. 7) identify three key findings from their study: GenAI requires entrepreneurs to master diverse skills and engage in "*pre- and post-processing*" for sustained value; supporting communal experiences alleviates "*techno-anxieties*" during onboarding; and entrepreneurs creatively use different tools, navigating concerns such as usability and efficiency. Their experiences are shaped by "*pre- and post-processing*," "*techno-anxieties*," and "*initial concerns.*" (Kotturi et al., 2024, p. 7). Together, these studies suggest that AI effectiveness depends on both its technical capabilities and the context of its implementation.

Ethical and contextual concerns are recurring themes in discussions about GenAI. Saleh and Semaan (2024) and Tran and Murphy (2023) caution against overreliance on these technologies, highlighting risks such as biases, privacy breaches, and data misuse. Saleh and Semaan (2024) focus on AI's potential in financial sentiment analysis, highlighting the role of prompt engineering in financial sentiment analysis, particularly zero-shot contexts, and compares ChatGPT's performance to FinBERT. While ChatGPT provides sentiment analysis, FinBERT delivers more accurate results for positive financial sentiments. These insights emphasize ChatGPT's potential for general analysis while highlighting FinBERT's superiority in domain-specific tasks like entrepreneurial sustainability (Saleh & Semaan, 2024). Tran and Murphy (2023) similarly recognize AI's ability to democratize entrepreneurial resources and bridge expertise gaps to reduce inequities but emphasize the need for critical and ethical application. They warn that "*the essential logic of generative AI draws from what is common, which can clash with the entrepreneurial mindset*," adding that its misuse could enable "*financial misinformation, illegal activities, or plagiarism*" (Tran & Murphy, 2023, p. 855).

Similarly, Norbäck and Persson (2024) provide a broader economic perspective, analyzing how GenAI reshapes market dynamics. They highlight its ability to enhance the competitive advantages of established players while motivating riskier innovations among new entrants. Their study found that "*limited access to operational data can induce entrepreneurs to take on more risk*," increasing the likelihood of transformative products (Norbäck & Persson, 2024, p. 371). They further note that increased machine learning applied to incumbents' operational data could make the process of creative destruction "*more creative but also less destructive*." (Norbäck & Persson, 2024, p. 371).

Finally, Reichert et al. (2024) explore the integration of Large Language Models (LLMs) within a serious game designed to empower female entrepreneurs by fostering resilience and raising awareness of gender biases. The LLM supports the creation of interactive visual novels, detects bias, and provides personalized feedback, encouraging reflection on real-world experiences. Ethical considerations including data privacy, inclusivity, and the emotional challenges of addressing discrimination, are thoughtfully addressed. The authors stress the importance of vigilant monitoring to prevent the reinforcement of biases or inaccuracies in LLM outputs. By emphasizing inclusivity and ethical data practices, the study highlights LLMs' potential in socially responsible applications (Reichert et al., 2024).

While Reichert et al. (2024) focus on AI-driven tools for addressing societal biases and empowering female entrepreneurs, Wang and Zhang (2024) take a broader perspective by examining systemic inequities within green enterprises and green entrepreneurship. Wang and Zhang (2024) emphasize the importance of policy interventions, such as financial incentives and training programs, to ensure equitable access to GenAI technologies and promote green entrepreneurship. Together, these studies underscore the necessity of integrating ethical considerations into AI systems to harness their transformative potential while mitigating unintended societal impacts. The convergence of these findings highlights the dual responsibility of leveraging AI's capabilities and safeguarding against its social consequences.

Discussions and research agenda

The content from "Technology adoption and integration" to "Performance and productivity enhancement" sections highlights the transformative potential of GenAI (GenAI) in entrepreneurship, intertwining themes of technology adoption, skill development, innovation, and performance enhancement. An analysis across different sections reveals the interconnected nature of these dimensions, synthesizing a holistic perspective on the challenges and opportunities GenAI offers.

A recurring theme is the essential role of readiness and support systems in integrating GenAI into entrepreneurial practices. Adoption pathways are shaped by psychological factors, such as self-efficacy and technostress, alongside external enablers like incubators and accessible resources (Duong & Nguyen, 2024). These psychological and structural elements resonate across discussions on skill development and performance enhancement, emphasizing the need for technical infrastructure and human-centric strategies to build confidence and reduce stress. For example, while "Technology adoption and integration" and "Performance and productivity enhancement" sections emphasize tangible and intangible resources, the educational focus in "Entrepreneurship education, research and skill development" section highlights the importance of equipping entrepreneurs with AI-related competencies through experiential learning for enhancing their self-efficacy and critical thinking abilities (Park & Sung, 2023; Sudirman & Rahmatillah, 2023; Winkler et al., 2023). These insights suggest that readiness encompasses not only access but also mindset and capability.

The integration of GenAI into education and innovation demonstrates its dual role as a learning tool and as a strategic enhancer. AI-driven pedagogies enable personalized and iterative learning experiences, fostering entrepreneurial skills like storytelling, creativity, collaboration, and strategic decision-making (George-Reyes et al., 2024; Liang & Bai, 2024). These educational advancements complement innovation efforts, where AI facilitates structured ideation and storytelling. The emphasis on skill-building in education directly translates into innovation outcomes, as tools like ChatGPT empower entrepreneurs to experiment with business models, engage stakeholders, and refine strategic narratives (Gaertner et al., 2023; Short & Short, 2023). This alignment illustrates that educational efforts are foundational to practical applications, enabling real-world problem-solving and innovation.

GenAI's role in performance enhancement and productivity builds on the preparatory themes of adoption and education while shifting focus to operational impact. The adaptability and efficiency of GenAI emerge as key drivers of improved entrepreneurial outcomes (Liu & Wang, 2024; Wang & Zhang, 2024). However, these productivity gains depend on the same factors that influence adoption-technological readiness, ability to address ethical challenges, psychological preparedness, and the presence of support systems (Kotturi et al., 2024; Saygin et al., 2024). Furthermore, performance discussions intersect with ethical considerations, emphasizing that productivity should not come at the expense of inclusivity or fairness. For instance, "Innovation and business model development" and "Performance and productivity enhancement" sections highlight the democratizing potential of GenAI research focussed on bridging barriers for marginalized entrepreneurs while cautioning against risks like biases and over-reliance on AI (Jiang et al., 2024; Wang & Zhang, 2024). This interplay underscores the importance of pursuing productivity within an ethical framework that prioritizes equity and transparency.

Ethical considerations permeate all themes, serving as a critical lens for evaluating adoption, education, innovation, and performance. Biases, privacy concerns, and unequal access repeatedly surface, emphasizing the dual responsibility of harnessing AI's capabilities while mitigating unintended consequences (Ilagan et al., 2024; Reichert et al., 2024). For example, while innovation discussions celebrate AI's role in enhancing communication and stakeholder engagement, they also caution against reduced human collaboration and creativity (Gaertner et al., 2023). Similarly, adoption challenges highlight disparities in readiness and access, framing equity as essential to ethical AI use (Etemad, 2024; Gupta, 2024). These intersections demonstrate that ethics are not peripheral concerns but foundational to shaping the deployment and impact of GenAI across contexts.

The relationship between human and technological elements is another unifying thread. GenAI serves as both an enabler of entrepreneurship and a catalyst for reshaping ecosystems. However, its transformative potential depends on human oversight and critical engagement. This theme is evident in skill development, where AI is most effective when paired with traditional methodologies to foster critical thinking and deep learning (Bell & Bell, 2023; Vecchiarini & Somià, 2023). Similarly, innovation and performance discussions highlight that AI's iterative capabilities require human evaluation to ensure relevance and effectiveness (Shin et al., 2024). Across sections, the synthesis underscores that GenAI's success lies in its ability to comple-

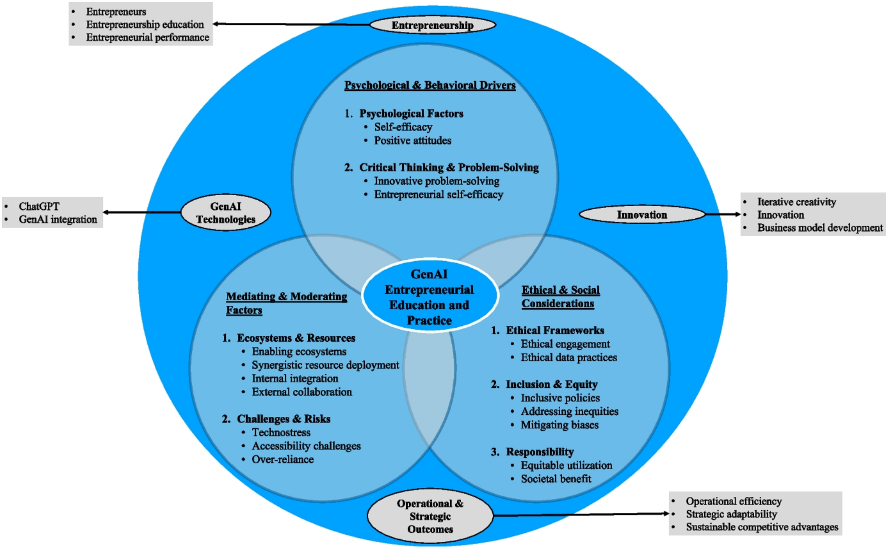

Fig. 1 The GAIN (The acronym GAIN stands for Guiding (steering responsible GenAI adoption), AI Integration (focusing on integrating Generative AI), Innovation & Impact (fostering entrepreneurial innovation and measurable outcomes), and Nurturing Ecosystems (highlighting the importance of enabling environments and ethical considerations). It reflects the framework's focus on responsible AI adoption, innovation, and ecosystem development in entrepreneurship.) Framework: Guiding AI Integration for Nurturing Entrepreneurship

ment, not replace, human ingenuity, emphasizing the importance of balanced, hybrid approaches.

In synthesizing these themes, it is clear that GenAI is not a panacea but a powerful enabler and its benefits and risks are deeply intertwined. Its adoption and integration depend on psychological, educational, operational, and ethical factors, each influencing the other. The holistic potential of GenAI is best realized through a systemic approach that aligns readiness, skill-building, innovation, and performance with robust ethical safeguards. Stakeholders must critically engage with AI, fostering ecosystems that prioritize inclusivity, transparency, and adaptability (Wang & Zhang, 2024). By addressing these interconnected dimensions, GenAI can serve as a transformative force for entrepreneurship, driving innovation and growth while upholding equity and responsibility.

Based on the discussion presented in the preceding parts of this paper, the GAIN Framework: Guiding AI Integration for Nurturing Entrepreneurship, depicted in Fig. 1, illustrates the psychological, operational, and ethical-societal dimensions of GenAI integration in entrepreneurship. Moreover, it is important that the framework reflects critical factors—such as self-efficacy, enabling ecosystems, and ethical considerations—to optimize GenAI adoption and sustain competitive advantage. By aligning technological integration with entrepreneurial education and inclusive practices, the framework aims to support responsible and equitable GenAI-driven outcomes, which are essential for ensuring that the evolving entrepreneurial landscape fosters mindful innovation, efficiency, and adaptability. The framework thus comprises the following three key overlapping areas:

Psychological and behavioural drivers- here, psychological factors, critical thinking, and problem-solving are key in driving innovation and entrepreneurial action.

Mediating and moderating factors- this is where the ecosystem and resources come into play. The ecosystem needs to be enabling, with the appropriate and synergistic deployment of resources. Organizations cannot do this in isolation and need to consider the most appropriate ways to collaborate externally, while always being mindful of the challenges and risks posed by this technology.

Ethical and social considerations- GenAI adoption, regardless of the sector, needs to be carried out ethically, with clear frameworks and safety mechanisms (such as ethical data practices) implemented. Inclusion and equity also need to be embedded within policies to mitigate biases and address inequities, while also taking a socially responsible stance.

The framework above emphasises a balanced approach to GenAI adoption, leveraging psychological drivers and collaborative ecosystems while mitigating risks such as over-reliance and absent-minded use of this technology. Ethical and inclusive practices are critical to addressing biases and ensuring societal benefits, aligning GenAI integration with long-term sustainable development goals.

Entrepreneurs and educators must focus on fostering critical thinking, strategic adaptability, and equitable utilisation to maximize the potential of GenAI in innovation and business model development. By operationalising these insights, the framework supports a path toward sustained innovation, enhanced entrepreneurial performance, and responsible technological transformation that allows for strategic outcomes.

Based on the intersection of the three overlapping areas illustrated in Fig. 1, five research propositions are outlined in "Research agenda" sub-section to hypothesize the interplay of psychological drivers, enabling ecosystems, and ethical imperatives in optimizing GenAI's impact on entrepreneurial education and practice. These propositions support a balanced approach to GenAI adoption, emphasizing entrepreneur empowerment, risk mitigation, and the promotion of equitable and sustainable practices.

Research agenda

Building on the themes synthesized in the preceding discussions and as illustrated in Fig. 1, this subsection highlights key areas where GenAI intersects with entrepreneurial practice, addressing both its transformative potential and challenges. By focusing on adoption, education, innovation, performance, and ethics, the following research propositions offer specific directions for exploring GenAI's role in shaping the future of entrepreneurship.

The adoption and integration of GenAI tools like ChatGPT among entrepreneurs represent a dynamic interplay of technological capabilities, individual cognitive factors, and socio-economic contexts. Entrepreneurs' perceptions of ChatGPT's usefulness and ease of use (Gupta, 2024), supported by social norms and prior technology experience, drive initial experimentation and foster confidence in the technology (Duong, 2024). Frameworks like the Theory of Planned Behavior and Social Cognitive Career Theory emphasize that self-efficacy and subjective norms are critical in shaping attitudes and intentions toward adoption (Bui & Duong, 2024; Duong, 2024). Moreover, ChatGPT acts as a resource for decision-making and market insights, enhancing digital entrepreneurial intentions and fostering innovative solutions (Davidsson & Sufyan, 2023). However, challenges such as technostress and various disparities (economic and access related) highlight the need for supportive ecosystems (Bui & Duong, 2024; Goel & Nelson, 2024). These insights culminate in the following research proposition:

Research Proposition 1: *Entrepreneurs' successful adoption and integration of GenAI technologies like ChatGPT are optimized when psychological drivers such as self-efficacy and positive attitudes, mediated by enabling ecosystems and moderated by contextual challenges like technostress and accessibility, are aligned to foster sustained innovation and operational efficiency*.

GenAI tools like ChatGPT significantly enhance entrepreneurship education by fostering creative ideation, personalized learning, and practical application of entrepreneurial concepts (Park & Sung, 2023; Sudirman & Rahmatillah, 2023). These tools help students identify real-world problems, refine business models, and simulate entrepreneurial scenarios, promoting analytical and decision-making skills (Bell & Bell, 2023; Darnell & Gopalkrishnan, 2024; Vecchiarini & Somià, 2023). The Social Cognitive Career Theory framework highlights the role of AI in boosting entrepreneurial self-efficacy, a key predictor of entrepreneurial intentions (Bui & Duong, 2024). Despite challenges like technostress and over-reliance, thoughtful integration of AI into curricula complements traditional methodologies, preparing students for digitally driven entrepreneurship (Bell & Bell, 2023; Bui & Duong, 2024; Jung & Suh, 2024; Liang & Bai, 2024; Vecchiarini & Somià, 2023; Winkler et al., 2023). This discussion provides the basis for formulating the following proposition:

Research Proposition 2: *The integration of GenAI into entrepreneurship education fosters entrepreneurial self-efficacy and innovative problem-solving, contingent on structured implementation that balances GenAI-driven personalization with traditional pedagogical approaches, mitigating risks of over-reliance and promoting critical thinking*.

GenAI tools like ChatGPT significantly enhance innovation and business model development by facilitating iterative ideation, strategic foresight, and tailored decision-making processes (Kostis et al., 2024; Mueller-Saegebrecht & Lippert, 2024). These tools enable entrepreneurs and students to explore creative pathways, validate business strategies, and refine outputs through advanced features like SWOT analysis and customer segmentation (Mueller-Saegebrecht & Lippert, 2024). By integrating methodologies such as the TRIZ framework, ChatGPT bridges technical and market-oriented perspectives, supporting both experienced professionals and novices (Shin et al., 2024). However, risks like over-reliance and potential GenAI hallucinations necessitate critical engagement and ethical oversight (Dwivedi et al., 2024; Ferrati et al., 2024). Drawing from this discussion, the following proposition is proposed:

Research Proposition 3: *The integration of GenAI tools like ChatGPT into innovation and business model development fosters iterative creativity and strategic adaptability while requiring frameworks for ethical engagement and critical evaluation to mitigate risks of over-reliance and enhance entrepreneurial outcomes*.

GenAI technologies, such as ChatGPT, are redefining entrepreneurial performance by streamlining internal processes and enabling external collaboration (Liu & Wang, 2024). Tangible resources like advanced hardware and data support operational efficiency, while intangible factors, such as an innovative culture, enhance strategic alignment and adoption (Liu & Wang, 2024; Tran & Murphy, 2023). Human Resources play a crucial role by equipping teams with the skills to utilize GenAI effectively, fostering creativity and decision-making capabilities (Kotturi et al., 2024). Internal integration-optimization of workflows, teamwork, and decision-making and external collaboration-resource sharing and market expansion are critical mediators translating GenAI potential into competitive advantages (Ferrati et al., 2024; Liu & Wang, 2024). These advancements are complemented by ChatGPT's ability to analyze large datasets, self-learn from them, simulate decision-making, and generate tailored insights, supporting innovation and efficiency (Saygin et al., 2024; Wang & Zhang, 2024). This discussion serves as a foundation for the subsequent proposition:

Research Proposition 4: *The integration of GenAI technologies, when aligned with synergistic resource deployment and mediated by internal integration and external collaboration, significantly enhances entrepreneurial performance, fostering innovation and sustainable competitive advantages*.

GenAI technologies, such as ChatGPT, bring transformative potential to education and entrepreneurship while posing significant social implications (Ilagan et al., 2024; Reichert et al., 2024). The democratization of learning resources by GenAI offers opportunities for inclusivity but risks deepening the digital divide, particularly for underprivileged groups without adequate access to technology (Ilagan et al., 2024). Biases in AI training data may perpetuate social inequities, highlighting the need for domain-specific knowledge integration and robust oversight (Lang et al., 2024; Reichert et al., 2024). Ethical data practices, privacy concerns, bias in decision-making, misuse of personal data, transparency, and informed consent are critical to ensuring equitable and socially responsible AI deployment (Saygin et al., 2024; Thanasi-Boçe & Hoxha, 2024; Tran & Murphy, 2023). Based on this discussion, the following proposition is put forward:

Research Proposition 5: *The social implications of GenAI integration in entrepreneurship require addressing inequities through inclusive policies, ethical data practices, and proactive measures to mitigate biases, ensuring equitable and responsible utilization for societal benefit*.

Conclusions

This review article aims to analyze early literature exploring issues related to GenAI and entrepreneurship. The findings reveal that GenAI is reshaping entrepreneurship by integrating technology, education, and ethics to drive innovation, efficiency, and inclusivity. Across technology adoption, skill development, and business performance, GenAI demonstrates immense potential to empower entrepreneurs by streamlining operations, enhancing decision-making, and democratizing access to resources. However, realizing these benefits requires addressing challenges such as ethical data use, biases, and equitable access to prevent technological advancements from exacerbating existing inequalities. The synergy between human creativity and GenAI-driven capabilities emerges as a crucial factor for success, emphasizing the need for balanced integration and robust safeguards. A central takeaway is the necessity of maintaining equilibrium between human ingenuity and AI capabilities, ensuring that AI serves as a complementary tool rather than as a replacement for critical thinking, creativity, and authentic learning. By integrating AI intentionally and methodically within structured educational frameworks, its potential can be harnessed while preserving the rigor, autonomy, and depth of human-driven learning and innovation. As these transformative tools continue to evolve, pressing questions arise: *How can entrepreneurs harness GenAI to innovate responsibly while maintaining human oversight*? *How can GenAI systems be designed or adapted to support entrepreneurial inclusivity and foster ethical business practices in a competitive digital economy?*

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References

Abdelwahed, N. A. A. (2024). Recognizing the role of ChatGPT in decision-making and recognition of mental health disorders among entrepreneurs. *OBM Neurobiology, 8*(3), 241. https://doi.org/10.21926/obm.neurobiol.2403241

Al-Busaidi, A. S., Raman, R., Hughes, L., Albashrawi, M. A., Malik, T., Dwivedi, Y. K., ... & Walton, P. (2024). Redefining boundaries in innovation and knowledge domains: Investigating the impact of generative artificial intelligence on copyright and intellectual property rights. *Journal of Innovation & Knowledge, 9*(4), 100630.

Bell, R., & Bell, H. (2023). Entrepreneurship education in the era of generative artificial intelligence. *Entrepreneurship Education, 6*(3), 229–244. https://doi.org/10.1007/s41959-023-00099-x

Bui, H. N., & Duong, C. D. (2024). ChatGPT adoption in entrepreneurship and digital entrepreneurial intention: A moderated mediation model of technostress and digital entrepreneurial self-efficacy. *Equilibrium. Quarterly Journal of Economics and Economic Policy, 19*(2), 391–428. https://doi.org/10.24136/eq.3074

Darnell, J. A., & Gopalkrishnan, S. (2024). Entrepreneurship teaching exercises: Integrating generative AI. *Discover Education, 3*(1), 172. https://doi.org/10.1007/s44217-024-00261-0

Davidsson, P., & Sufyan, M. (2023). What does AI think of AI as an external enabler (EE) of entrepreneurship? An assessment through and of the EE framework. *Journal of Business Venturing Insights, 20*, e00413.

Duong, C. (2024). ChatGPT adoption and digital entrepreneurial intentions: An empirical research based on the theory of planned behaviour. *Entrepreneurial Business and Economics Review, 12*(2), 129–142.

Duong, C. D., & Nguyen, T. H. (2024). How ChatGPT adoption stimulates digital entrepreneurship: A stimulus-organism-response perspective. *International Journal of Management Education, 22*(3), 101019. https://doi.org/10.1016/j.ijme.2024.101019

Duong, C. D., Nguyen, T. H., Chu, T. V., Pham, T. V., & Do, N. D. (2024). Whether ChatGPT adoption inspires higher education students' digital entrepreneurial intention? An integrated model of the SCCT and the TPB. *International Journal of Innovation Science*. Vol. ahead-of-print No. ahead-of-print. https://doi.org/10.1108/IJIS-01-2024-0020

Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., ... & Wright, R. (2023). Opinion Paper:"So what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management, 71*, 102642.

Dwivedi, Y. K., Malik, T., Hughes, L., & Albashrawi, M. A. (2024). Scholarly discourse on GenAI's impact on Academic Publishing. *Journal of Computer Information Systems*, 1–16. https://doi.org/10.1080/08874417.2024.2435386

Etemad, H. (2024). Transformative potentials of generative artificial intelligence: Should international entrepreneurial enterprises adopt GEN.AI? *Journal of International Entrepreneurship, 22*(2), 141–163. https://doi.org/10.1007/s10843-024-00363-8

Ferrati, F., Kim, P. H., & Muffatto, M. (2024). Generative AI in entrepreneurship research: Principles and practical guidance for intelligence augmentation. *Foundations and Trends® in Entrepreneurship, 20*(3), 245–383.

Gaertner, S., Patel, V., Cabera, D., Rafikia, E., Nkululeko, R., & Mehta, K. (2023). Using generative AI to strategize entrepreneurial endeavors. *2023 IEEE Global Humanitarian Technology Conference (GHTC 2023)* (pp. 250–251). IEEE. https://doi.org/10.1109/GHTC56179.2023.10354803

George-Reyes, C. E., Vilhunen, E., Avello-Martínez, R., & López-Caudana, E. (2024). Developing scientific entrepreneurship and complex thinking skills: Creating narrative scripts using ChatGPT. *Frontiers in Education, 9*, 1378564. https://doi.org/10.3389/feduc.2024.1378564

Goel, R. K., & Nelson, M. A. (2024). Awareness of artificial intelligence: Diffusion of AI versus ChatGPT information with implications for entrepreneurship. *Journal of Technology Transfer*. Advance online publication. https://doi.org/10.1007/s10961-024-10089-3

Gupta, V. (2024). An empirical evaluation of a generative artificial intelligence technology adoption model from entrepreneurs' perspectives. *Systems, 12*(3), 103. https://doi.org/10.3390/systems12030103

Herani, R., & Angela, J. (2024). Navigating ChatGPT: Catalyst or challenge for Indonesian youth in digital entrepreneurship? *Journal of Entrepreneurship in Emerging Economies*. Advance online publication. https://doi.org/10.1108/JEEE-05-2024-0181

Ilagan, J. B. R., Ilagan, J. R. S., & Rodrigo, M. M. T. (2024). Ethical education data mining framework for analyzing and evaluating large language model-based conversational intelligent tutoring systems for management and entrepreneurship courses. In *Lecture notes in networks and systems* (vol. 1011, pp. 61–71). https://doi.org/10.1007/978-981-97-4581-4_6

Jiang, Y., Jiang, Z., & Chen, Z. (2024). Women entrepreneurship in China: A bibliometric literature review and future research agenda. *Journal of Business Research, 179*, 114688.

Jung, D., & Suh, S. (2024). Enhancing soft skills through generative AI in sustainable fashion textile design education. *Sustainability, 16*(16), 6973.

Kasneci, E., Seßler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., ... & Kasneci, G. (2023). ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences, 103*, 102274.

Kostis, A., Lidstrom, J., Nair, S., & Holmstrom, J. (2024). Too much AI hype, too little emphasis on learning? Entrepreneurs designing business models through learning-by-conversing with generative AI. *IEEE Transactions on Engineering Management, 71*, 15278–15291. https://doi.org/10.1109/TEM.2024.3484750

Kotturi, Y., Anderson, A., Ford, G., Skirpan, M., & Bigham, J. P. (2024). Deconstructing the veneer of simplicity: Co-designing introductory generative AI workshops with local entrepreneurs. In *Conference on human factors in computing systems - proceedings* (Article 1014). https://doi.org/10.1145/3613904.3642191

Lang, Q., Tian, S., Wang, M., & Wang, J. (2024). Exploring the answering capability of large language models in addressing complex knowledge in entrepreneurship education. *IEEE Transactions on Learning Technologies*. https://doi.org/10.1109/TLT.2024.3456128

Liang, E. S., & Bai, S. (2024). Generative AI and the future of connectivist learning in higher education. *Journal of Asian Public Policy*, 1–23. https://doi.org/10.1080/17516234.2024.2386085

Liu, A., & Wang, S. (2024). Generative artificial intelligence (GenAI) and entrepreneurial performance: Implications for entrepreneurs. *Journal of Technology Transfer, 49*(6), 2389–2412. https://doi.org/10.1007/s10961-024-10132-3

Mueller-Saegebrecht, S., & Lippert, I. (2024). In tandem with ChatGPT-4: How LLM enhance entrepreneurship education and business model innovation. *Academy of Management Annual Meeting Proceedings, 2024*(1). https://doi.org/10.5465/AMPROC.2024.105bp

Norbäck, P. J., & Persson, L. (2024). Why generative AI can make creative destruction more creative but less destructive. *Small Business Economics, 63*(1), 349–377.

Park, J. Y., & Sung, C. S. (2023). The impact of generative AI tools on the development of entrepreneurial career intentions. In *Australasian conference on information systems* (ACIS 2023). Retrieved from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85200149145. Accessed 15 Nov 2024.

Reichert, T., Miftari, M., Herling, C., & Marsden, N. (2024). Empowering female founders with AI and Play: Integration of a large language model into a serious game with player-generated content. *International conference on human-computer interaction* (pp. 69–83). Springer Nature Switzerland.

Richey, R. G., Jr., Chowdhury, S., Davis-Sramek, B., Giannakis, M., & Dwivedi, Y. K. (2023). Artificial intelligence in logistics and supply chain management: A primer and roadmap for research. *Journal of Business Logistics, 44*(4), 532–549.

Saleh, L., & Semaan, S. (2024). The potential of AI in performing financial sentiment analysis for predicting entrepreneur survival. *Administrative Sciences, 14*(9), 220.

Saygin, M., Bekmezci, M., & Dinçer, E. (2024). Artificial intelligence model ChatGPT-4: Entrepreneur candidate and entrepreneurship example. *F1000Research, 13*, 308. https://doi.org/10.12688/f1000research.144671.2

Sedaghat, S. (2023). Early applications of ChatGPT in medical practice, education and research. *Clinical Medicine, 23*(3), 278–279.

Shin, W.-S., Lee, S.-H., & Sue, H.-J. (2024). Fostering tech innovation: Exploring TRIZ and ChatGPT integration for developer and entrepreneur challenges. *Tehnicki Glasnik, 18*(4), 588–597. https://doi.org/10.31803/tg-20231212081808

Shore, A., Tiwari, M., Tandon, P., & Foropon, C. (2024). Building entrepreneurial resilience during crisis using generative AI: An empirical study on SMEs. *Technovation, 135*, 103063. https://doi.org/10.1016/j.technovation.2024.103063

Short, C. E., & Short, J. C. (2023). The artificially intelligent entrepreneur: ChatGPT, prompt engineering, and entrepreneurial rhetoric creation. *Journal of Business Venturing Insights, 19*, e00388.

Somia, T., & Vecchiarini, M. (2024). Navigating the new frontier: The impact of artificial intelligence on students' entrepreneurial competencies. *International Journal of Entrepreneurial Behavior & Research, 30*(11), 236–260.

Sudirman, I. D., & Rahmatillah, I. (2023). Artificial intelligence-assisted discovery learning: An educational experience for entrepreneurship students using ChatGPT. *2023 IEEE World AI IoT Congress (AIIoT 2023)* (pp. 786–791). IEEE. https://doi.org/10.1109/AIIoT58121.2023.10174472

Thanasi-Boçe, M., & Hoxha, J. (2024). From ideas to ventures: Building entrepreneurship knowledge with LLM, prompt engineering, and conversational agents. *Education and Information Technologies*, 1–57. https://doi.org/10.1007/s10639-024-12775-z

Tran, H., & Murphy, P. J. (2023). Generative artificial intelligence and entrepreneurial performance. *Journal of Small Business and Enterprise Development, 30*(5), 853–856.

Vecchiarini, M., & Somià, T. (2023). Redefining entrepreneurship education in the age of artificial intelligence: An explorative analysis. *The International Journal of Management Education, 21*(3), 100879.

Wang, S., & Zhang, H. (2024). Green entrepreneurship success in the age of generative artificial intelligence: The interplay of technology adoption, knowledge management, and government support. *Technology in Society, 79*, 102744. https://doi.org/10.1016/j.techsoc.2024.102744

Winkler, C., Hammoda, B., Noyes, E., & Van Gelderen, M. (2023). Entrepreneurship education at the dawn of generative artificial intelligence. *Entrepreneurship Education and Pedagogy, 6*(4), 579–589. https://doi.org/10.1177/25151274231198799